

氮族元素答案

一. 小品打表 (36分, 占10%)

A组: 1. As_2S_3 2. AsH_3 3. Bi_2S_3 4. SbF_5 5. P_2H_4 6. P_4O_{10}

7. HN_3 8. NCl_3

B组: 1. H_2NOH 2. Mg_3N_2 3. NH_4Cl 4. NH_4NO_3 5. $H_4P_2O_7$ 6. Cu_3P

7. P_4O_6 8. $POCl_3$ 9. $GaAs$ 10. $SbOCl$

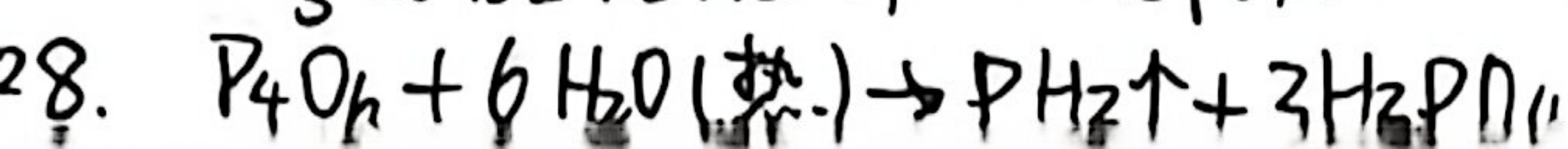
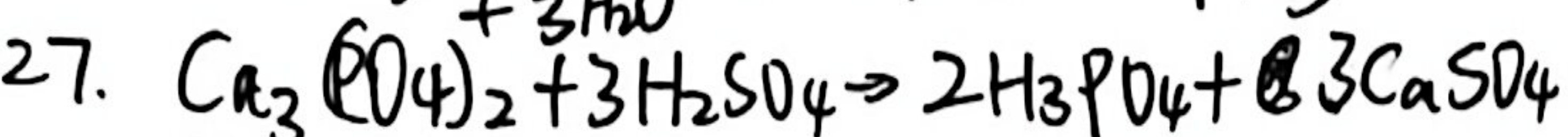
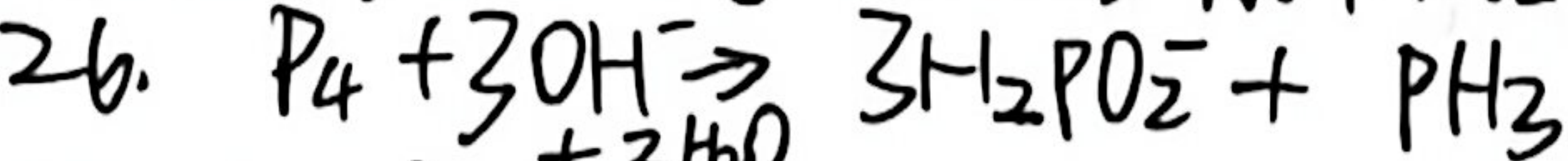
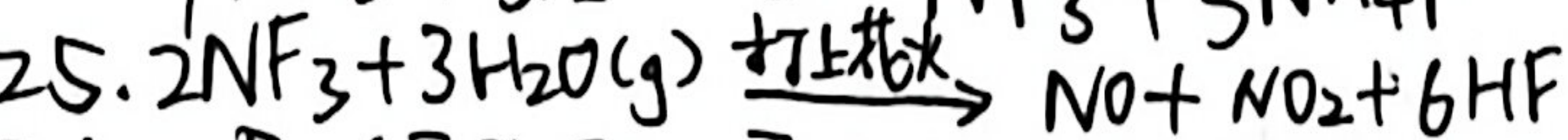
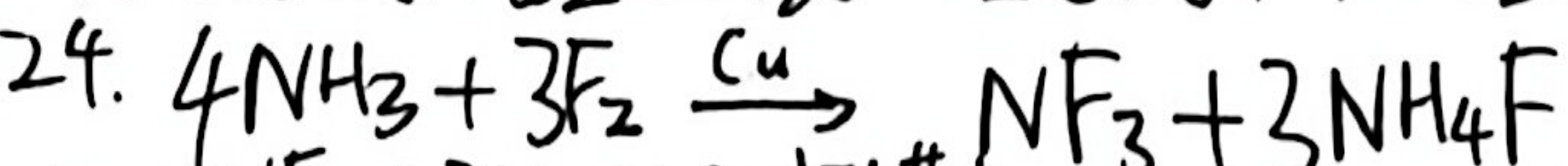
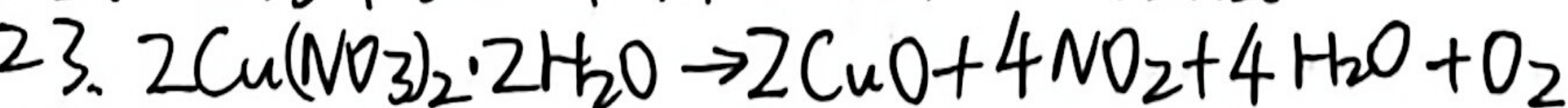
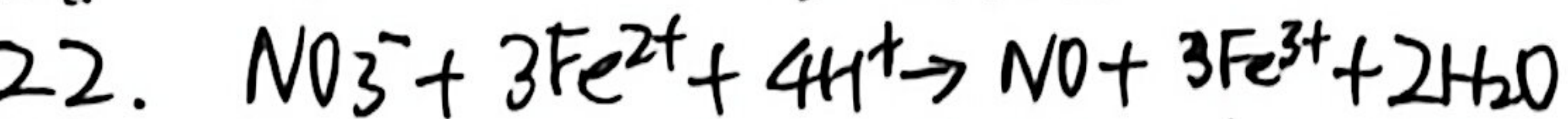
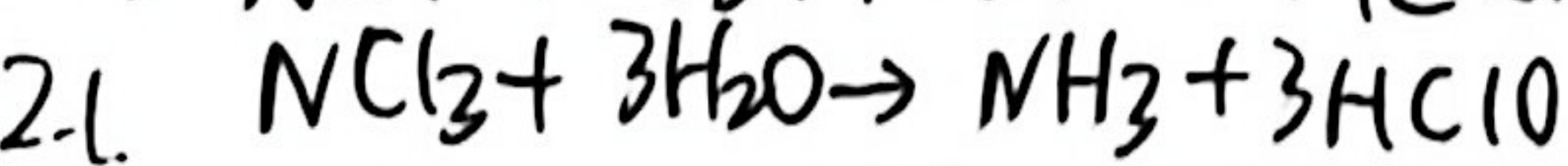
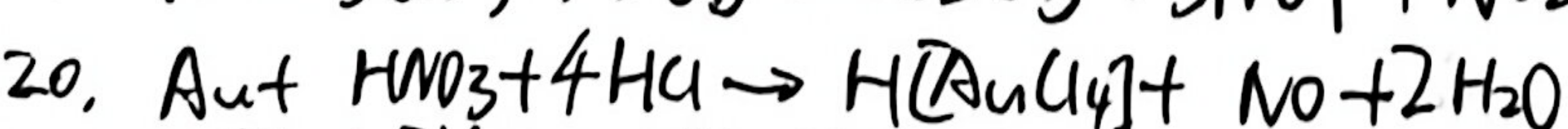
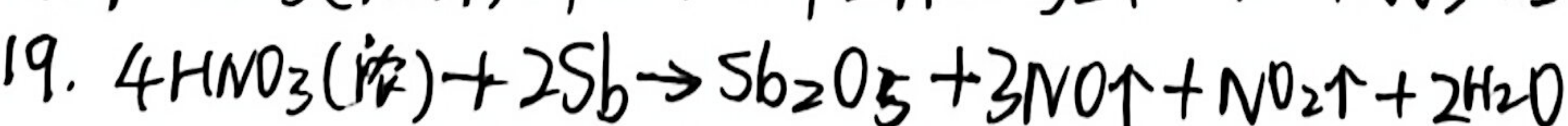
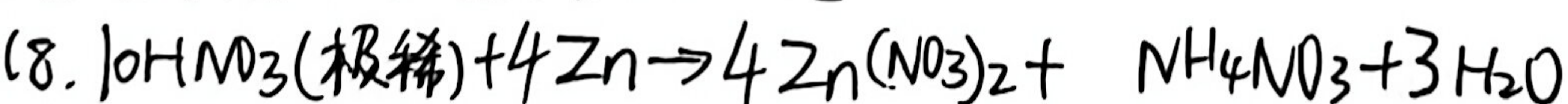
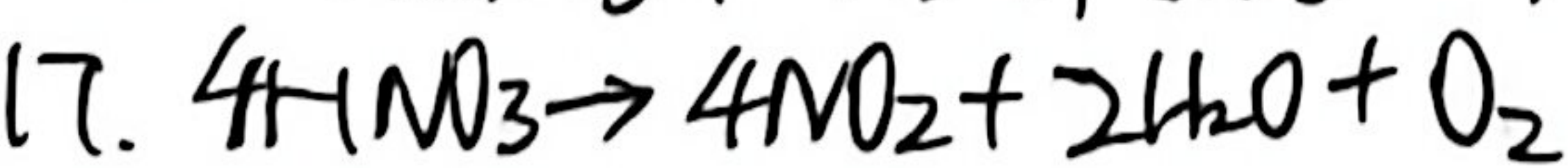
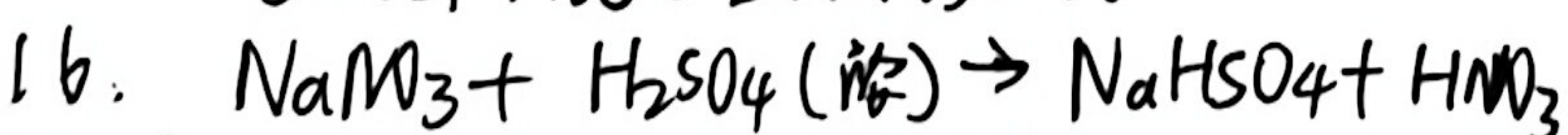
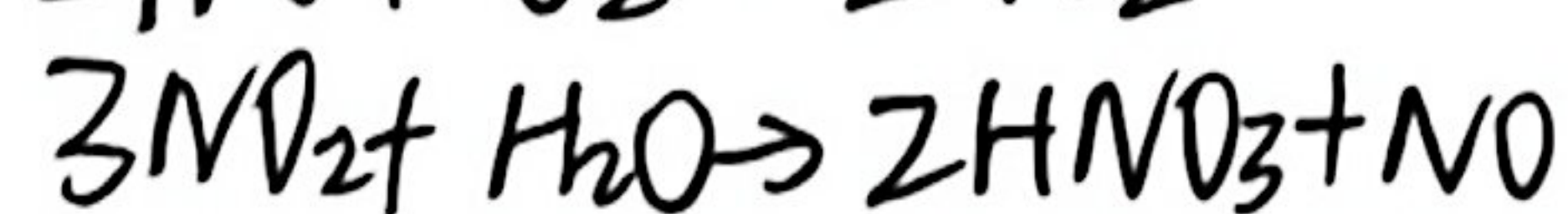
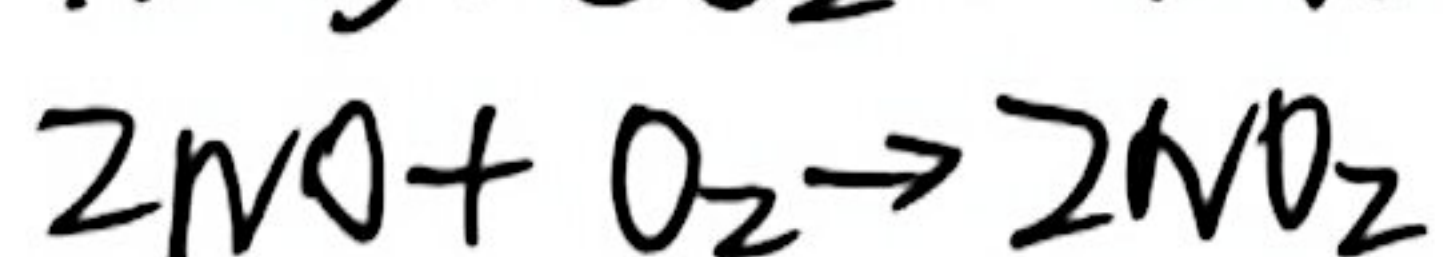
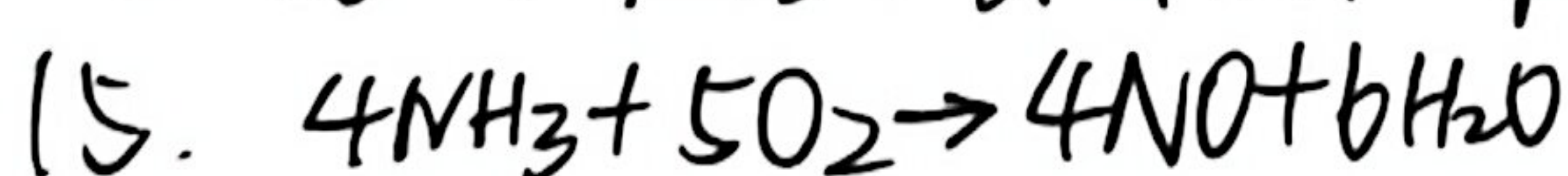
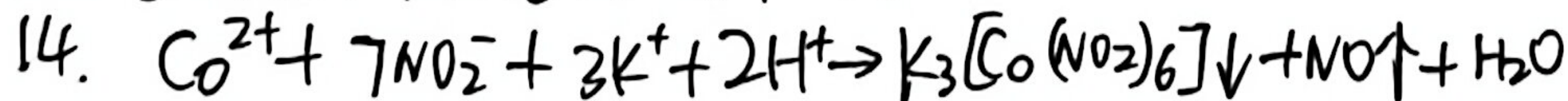
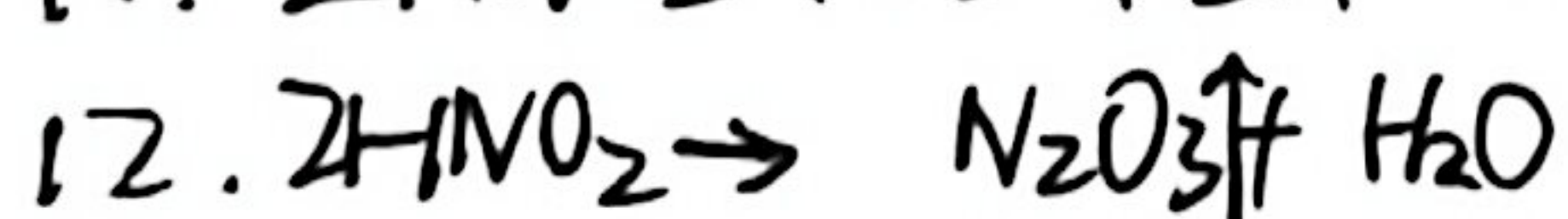
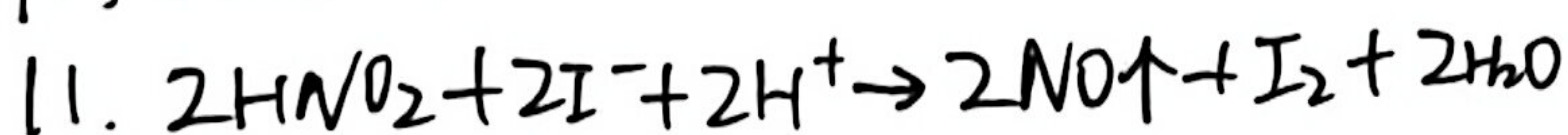
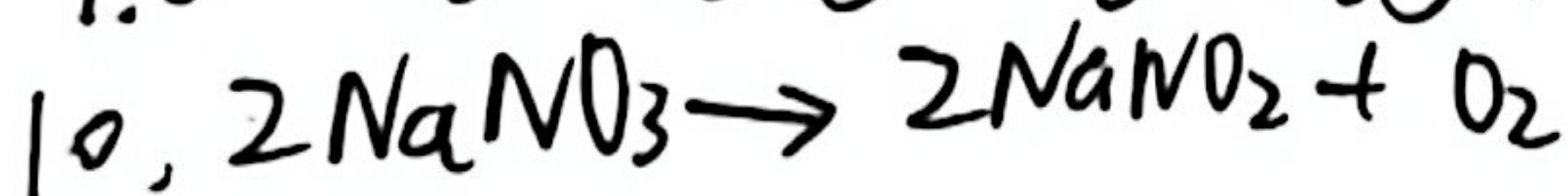
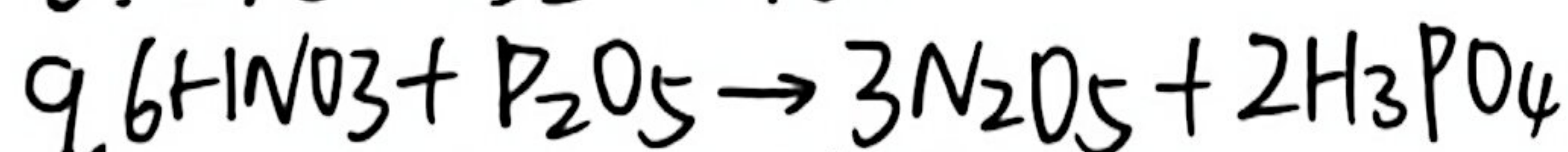
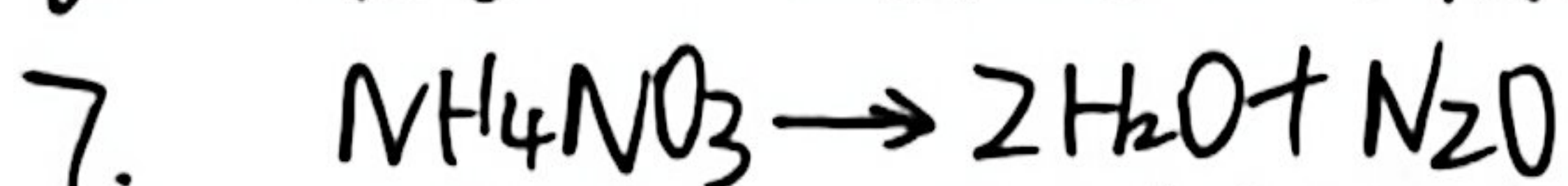
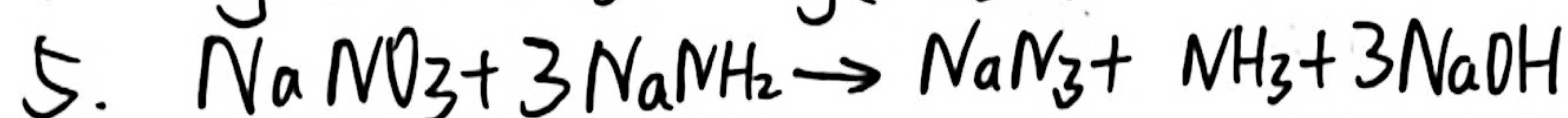
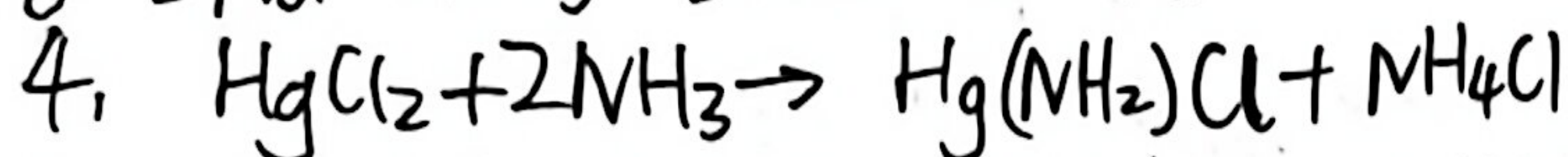
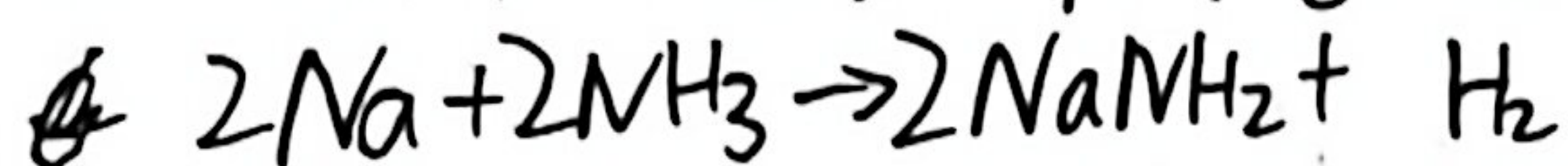
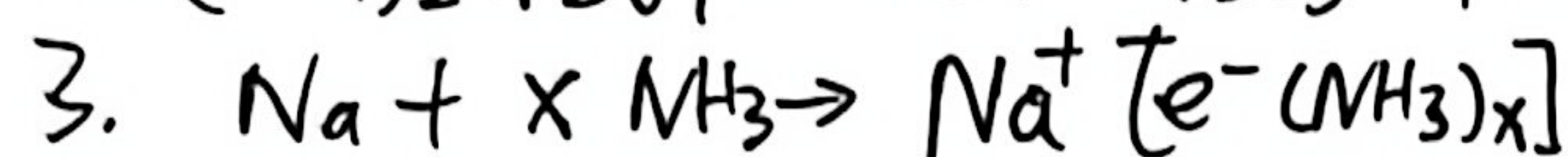
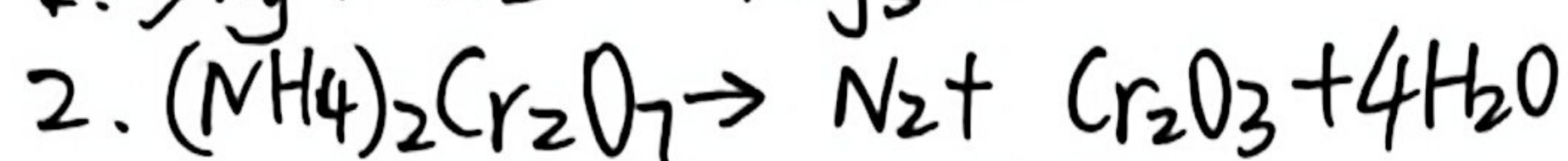
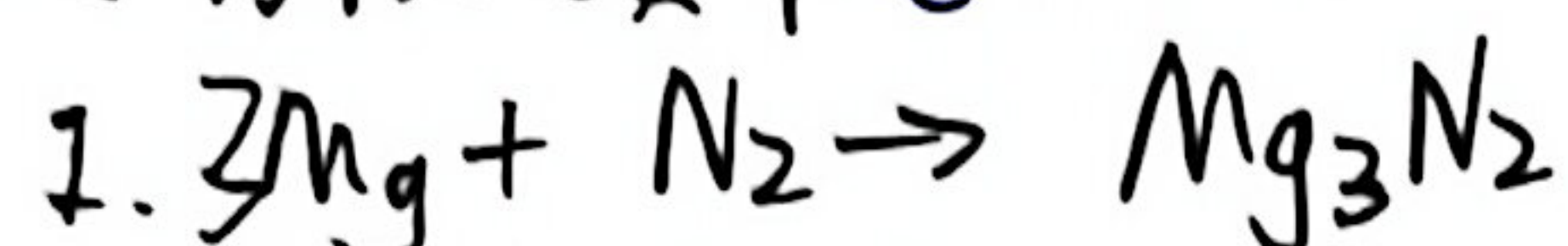
C组: 1. Fe_2P 2. P_4O 3. $Ca_5(PO_4)_3F$ 4. $NiSbS$ 5. $FeAs_2$ 6. $(NH_2)_2SO$

7. $Hg(NH_2)Cl$ 8. As_3H_5 9. $Pb(N_3)_2$ 10. $Tl[I(N_3)_4]$ 11. $K[OsO_3N]$

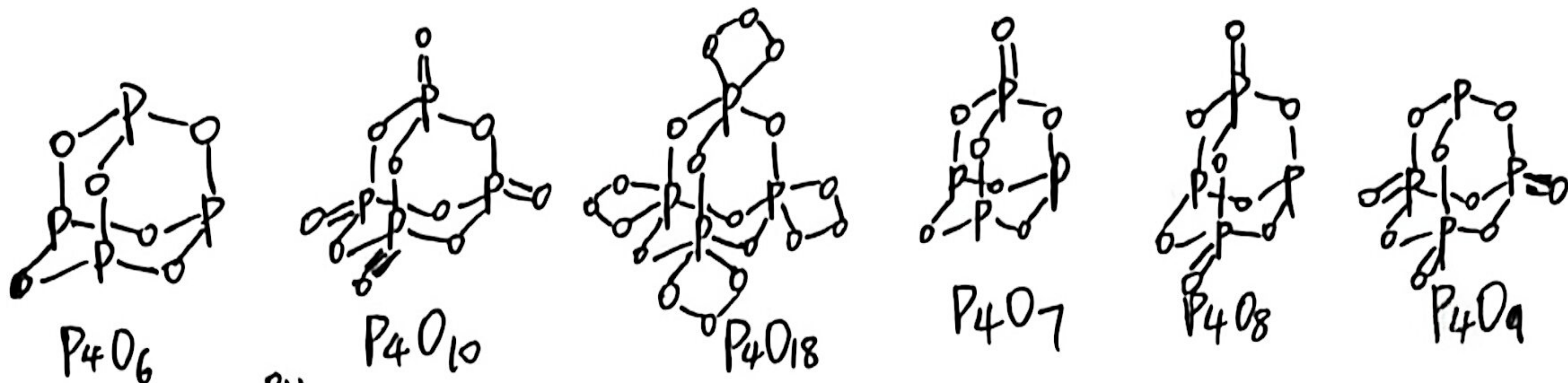
12. SF_5NF_2 13. $[Pt(N_2O_2)(PPh_3)_2]$ 14. $K[Au(NO_3)_4]$ 15. $Ca_8H_2(PO_4)_6 \cdot 5H_2O$

16. Th_3P_4 17. AsC_5H_5 18. Bi_6Cl_7

二. 方程式赛车 (30分, 占20%)

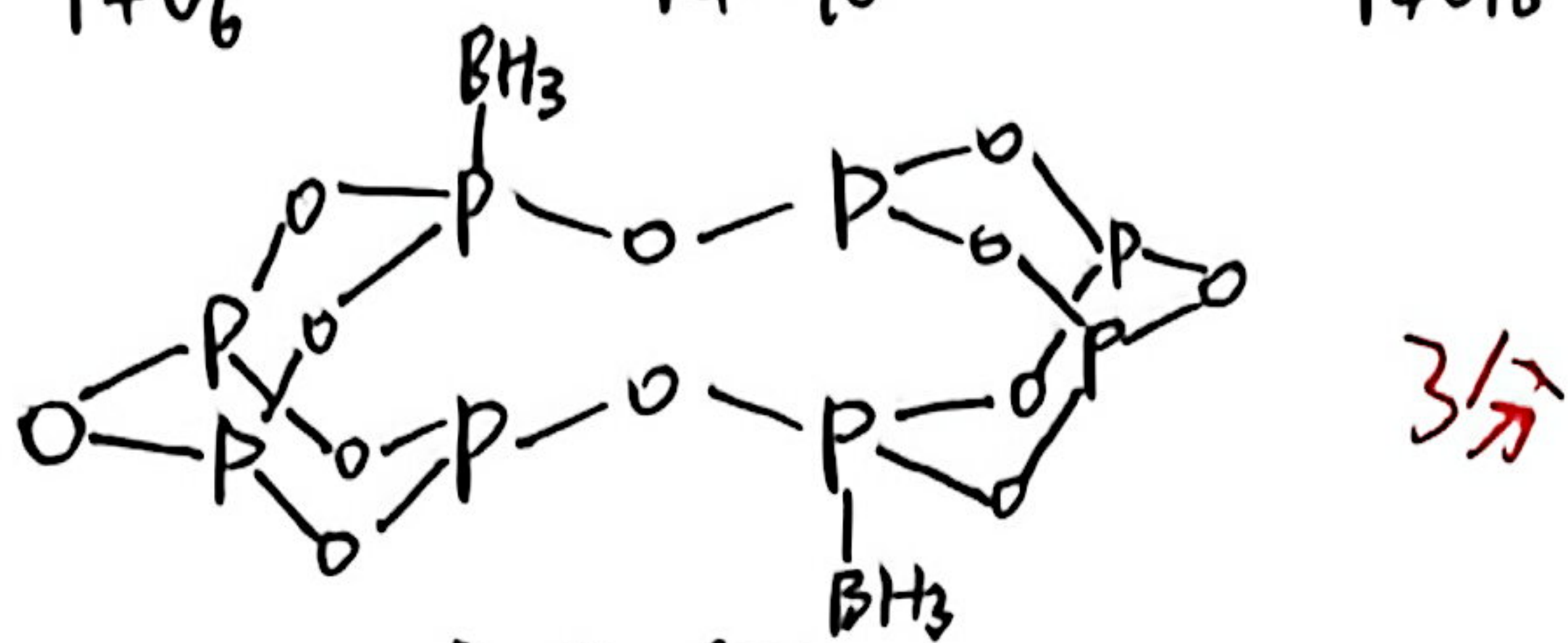


2.3.1



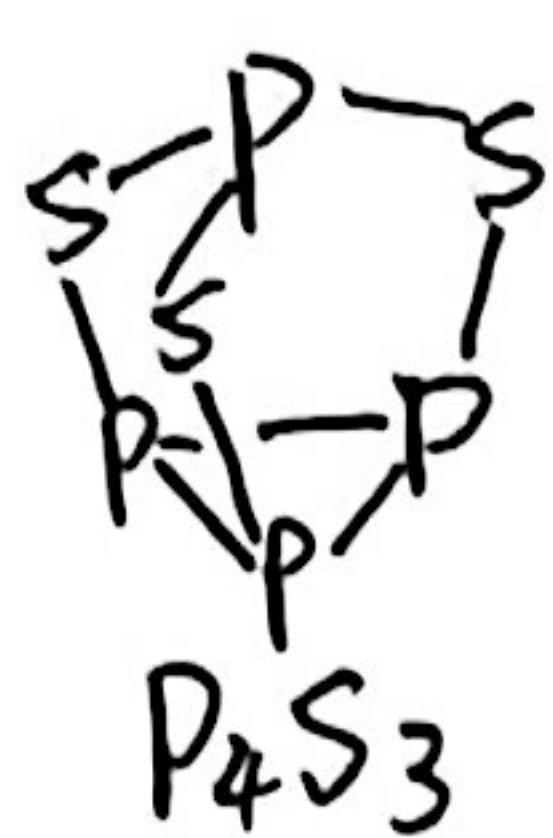
P_4O_{18} 2分, 其余各1分
共7分

2.3.2



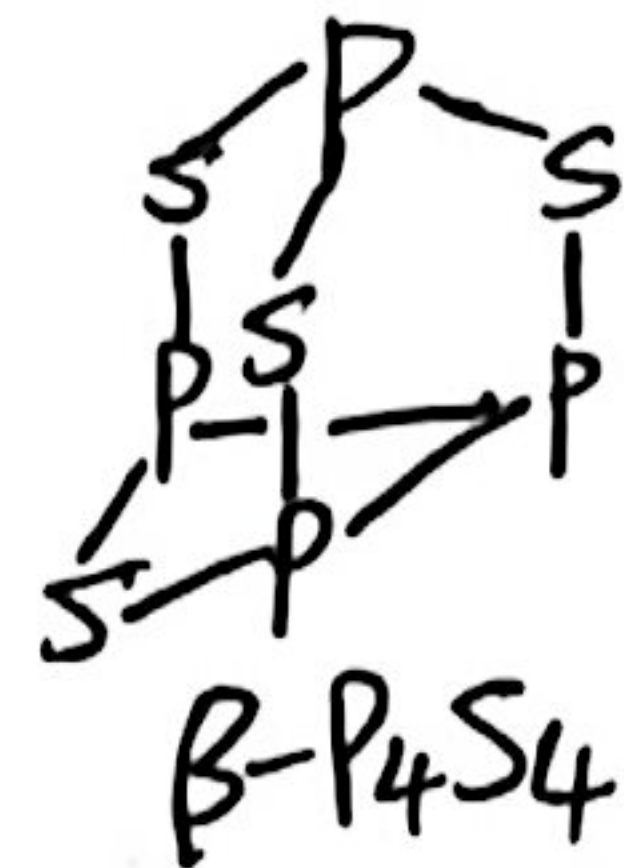
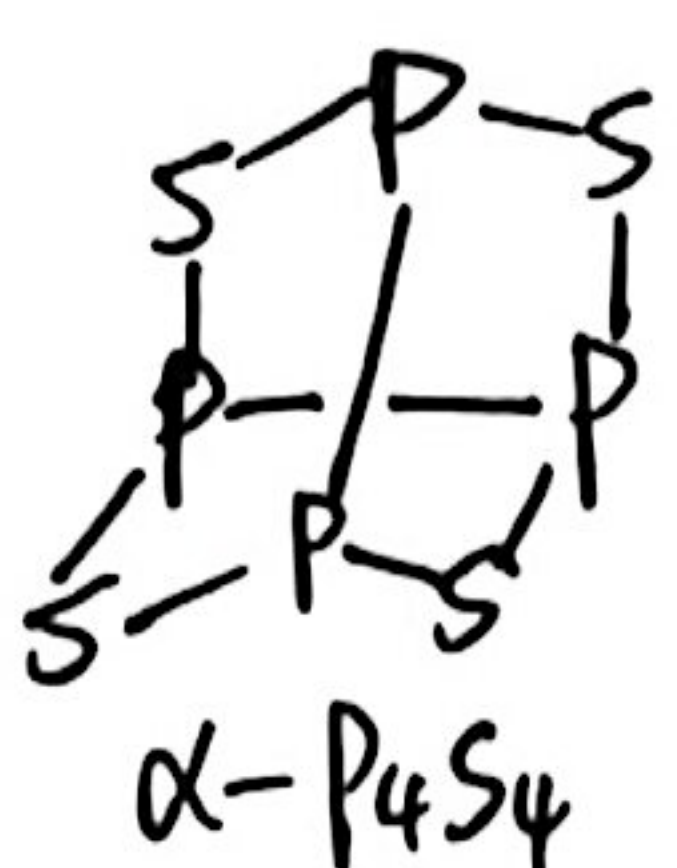
3分

2.3.3



2分

2.3.4

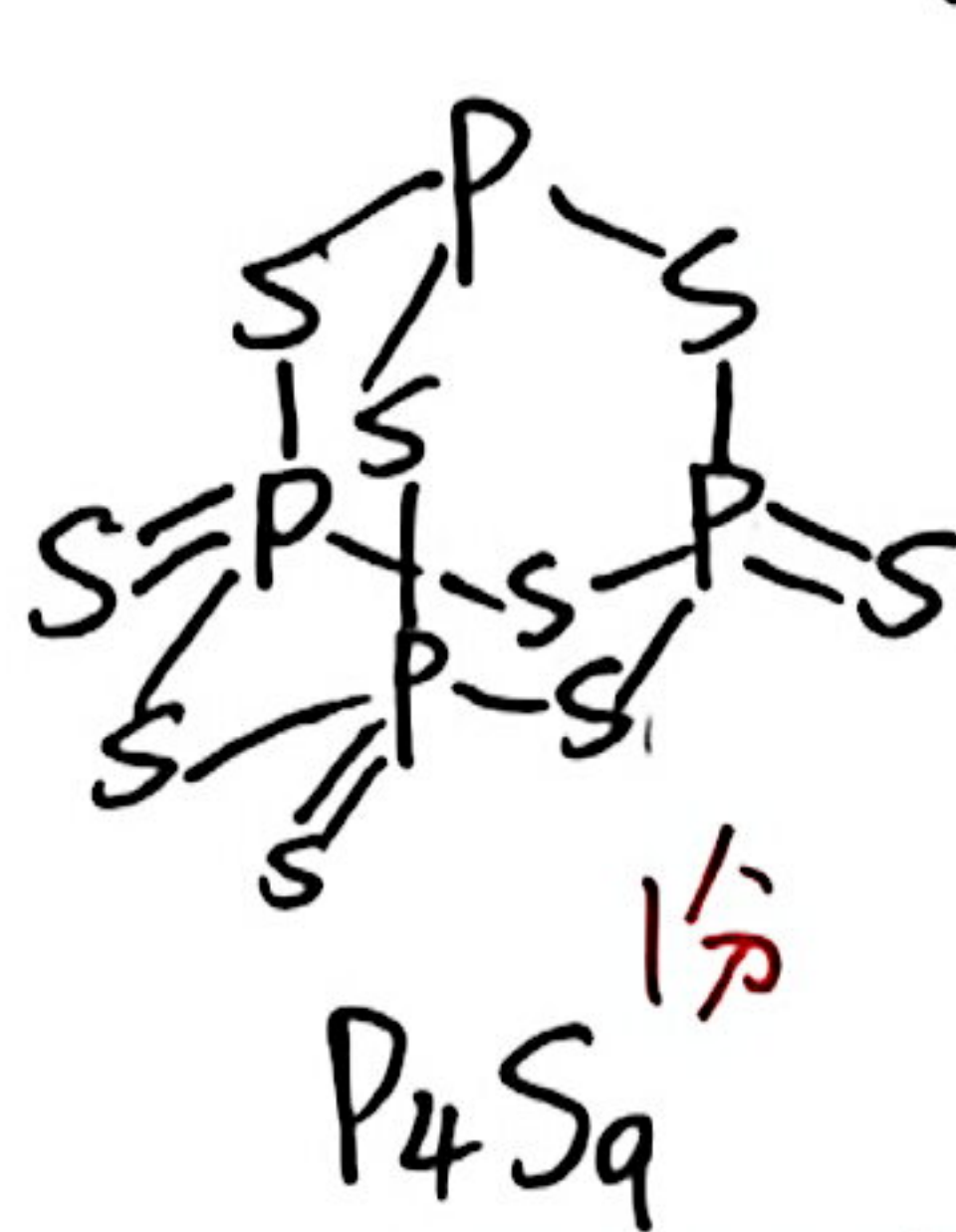


各2分, 共4分

2.3.5

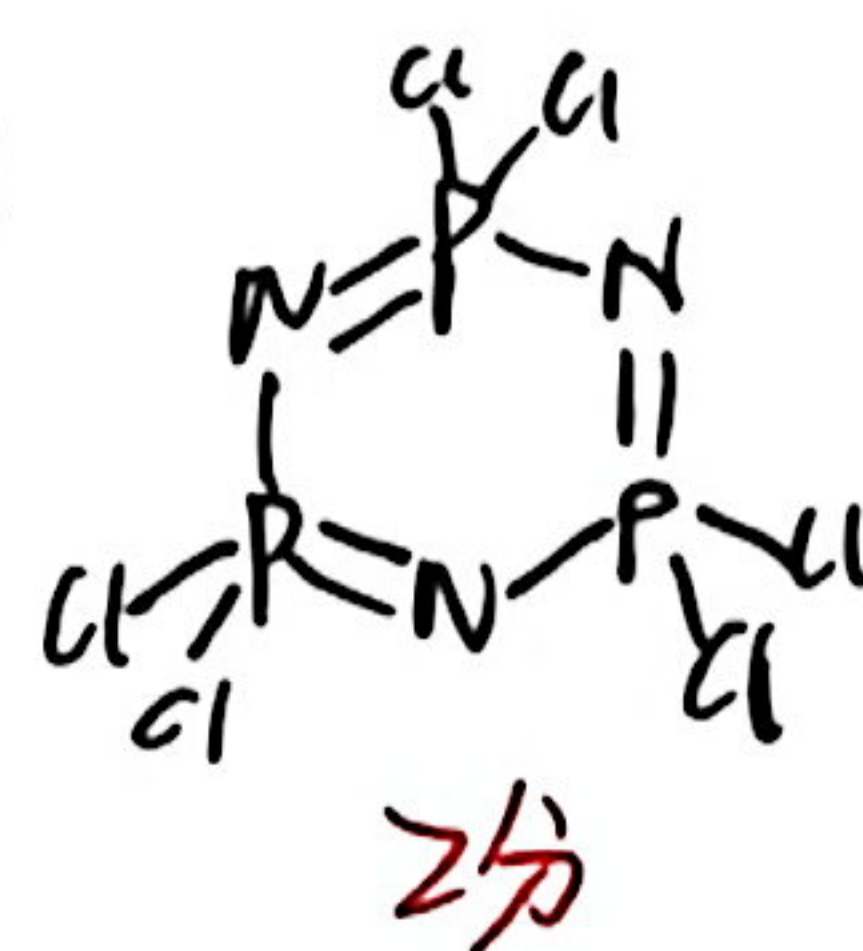


2分



1分

2.3.6.1



2.3.6.2 极易水解. 1分

第3题 危中! [注: 酸性条件下溶液中 As_2O_3 写成 H_3AsO_3 均给分] 34分

3.1 $e > a > c > b > d > f$ 3分 写出 a 和 c 毒性比 b 大占1分, b 比 d 毒性大占1分, 其余排对得满分

3.2 H 为 FeAsS 1分

反应1: $FeAsS + 3H_2O \rightarrow Fe(OH)_3 + H_3As + S$ 2分

反应2: $FeAsS \rightarrow FeS + As$ 2分

3.3.1 $Ag_3As \cdot 3AgNO_3$ 2分

3.3.2 反应3: $6Zn + 12HCl + As_2O_3 \rightarrow 2AsH_3 + 6ZnCl_2 + 3H_2O$ 2分

反应4: $2AsH_3 \rightarrow 2As + 3H_2$ 1分

反应5: $AsH_3 + 6AgNO_3 + 3H_2O \rightarrow 6Ag + 6HNO_3 + H_3AsO_3$ 2分

反应6: $AsH_3 + 6AgNO_3 + 3H_2O \rightarrow Ag_3As \cdot 3AgNO_3 + 3HNO_3$ 1分

反应7: $2As + 5ClO^- + 6OH^- \rightarrow 2AsO_4^{3-} + 5Cl^- + 3H_2O$ 2分

3.4. 反应8: $AsH_3 + 3HgBr_2 \rightarrow 3HBr + As(HgBr)_3$ 2分

反应9: $2As(HgBr)_3 + AsH_3 \rightarrow 3AsH(HgBr)_2$ 2分

反应10: $As(HgBr)_3 + AsH_3 \rightarrow 3HBr + As_2Hg_3$ 2分

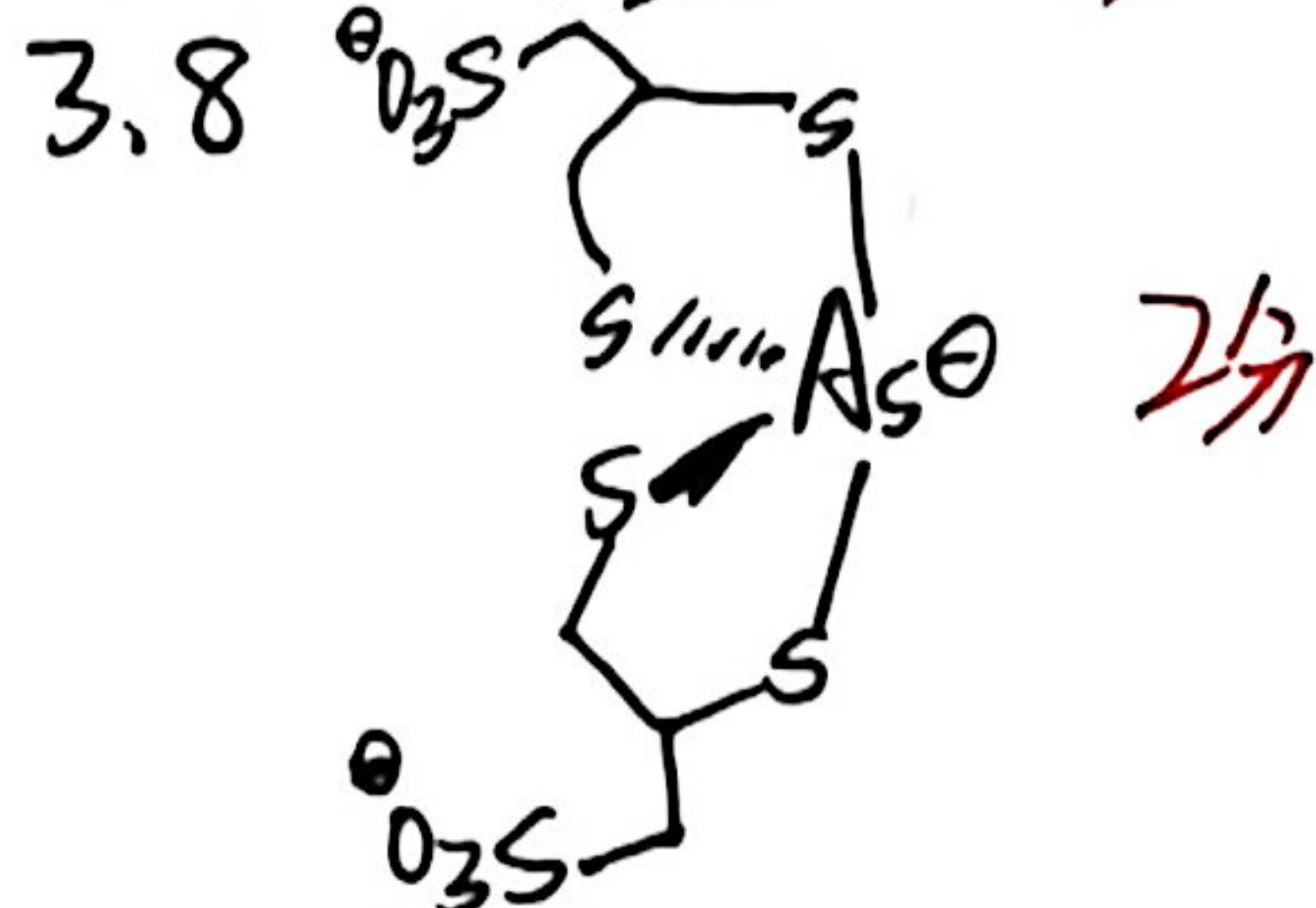
3.5 反应11: $AsH_3 + 6Ag(S_2CNEt_2) \rightarrow 6Ag + 3As(S_2CNEt_2)_3 + 3Et_2NCS_2H$ 2分

3.6 反应12: $2H_3AsO_3 + 3H_3PO_2 \rightarrow 3H_3PO_3 + 2As \downarrow + 3H_2O$ 2分

反应13: $3Sn^{2+} + 2As^{3+} \rightarrow 3Sn^{4+} + 2As$ 2分, Sn^{4+} 其他形式也可

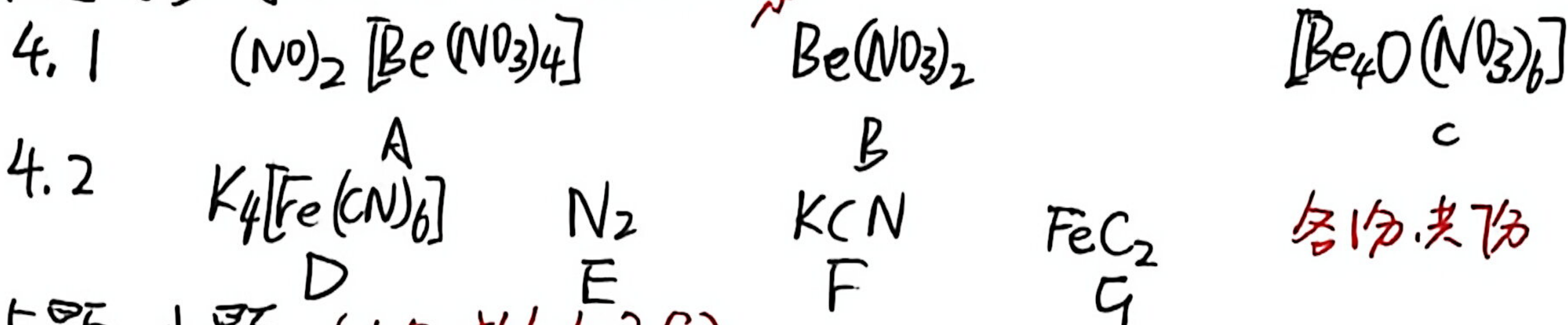
3.7.1 As(V) 毒性低 (海鲜中就含有 As(V) 如螃蟹), As 甚至为人体微量元素 1分

3.7.2 催吐 1分



五、千万次想象 (共5题, 折合加30分)

第4题 奇妙的热重 (7分, 折合加5分)



第5题 小题 (6分, 折合加3分)

5.1 ~~As~~ Sb 和 Bi 2分

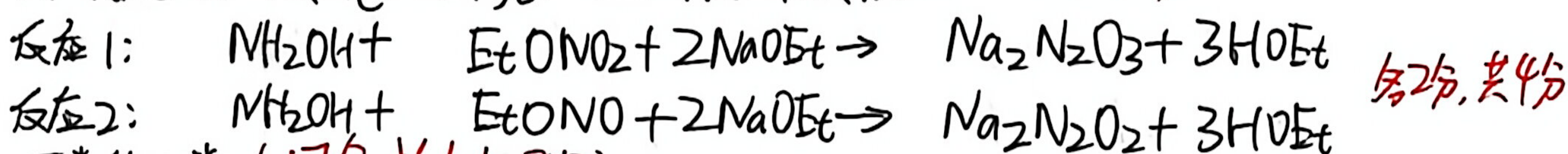
5.2 常共生: (b) 可共生: (c) 极少共生或显著不共生: (a)(d) 各1分, 共4分

第6题 "石硝"和它的朋友们 (9分, 折合加5分)

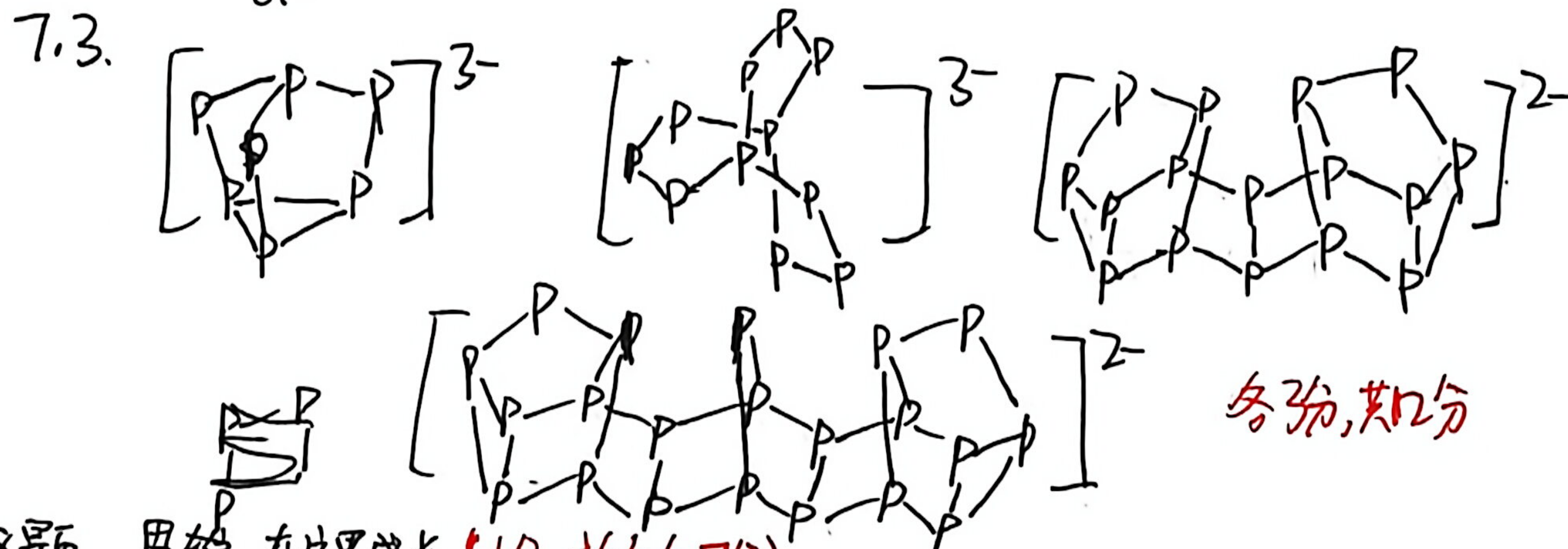
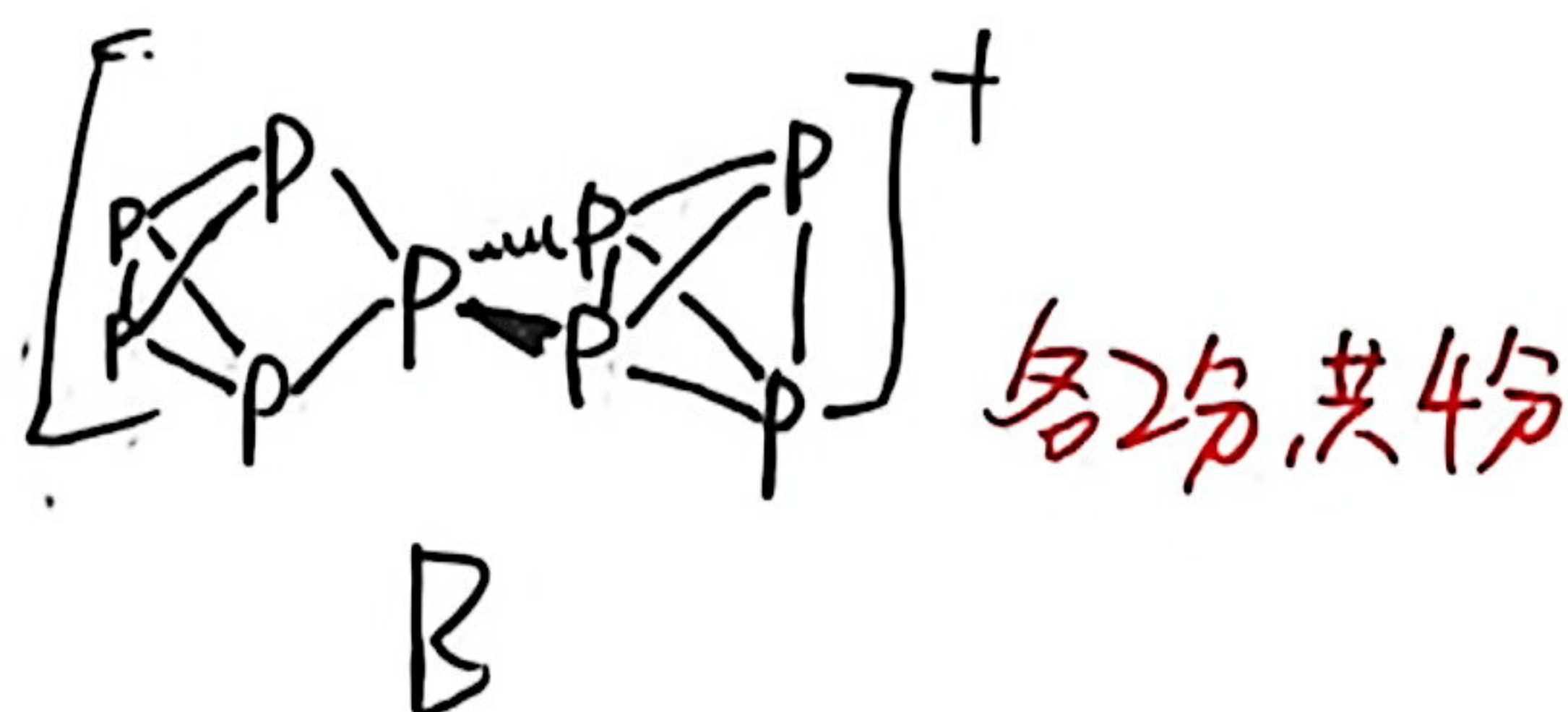
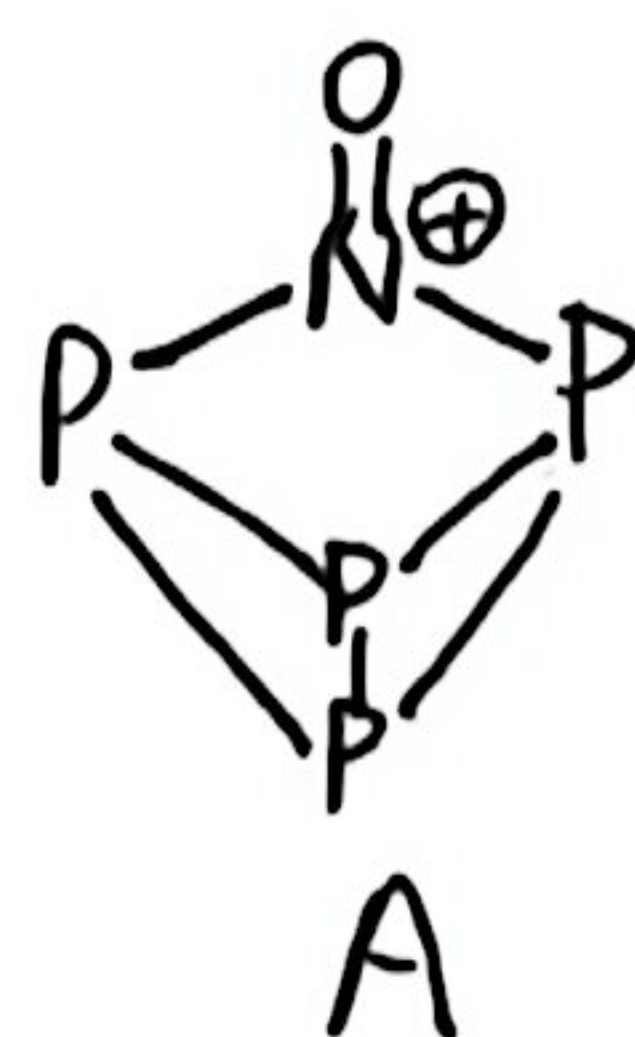
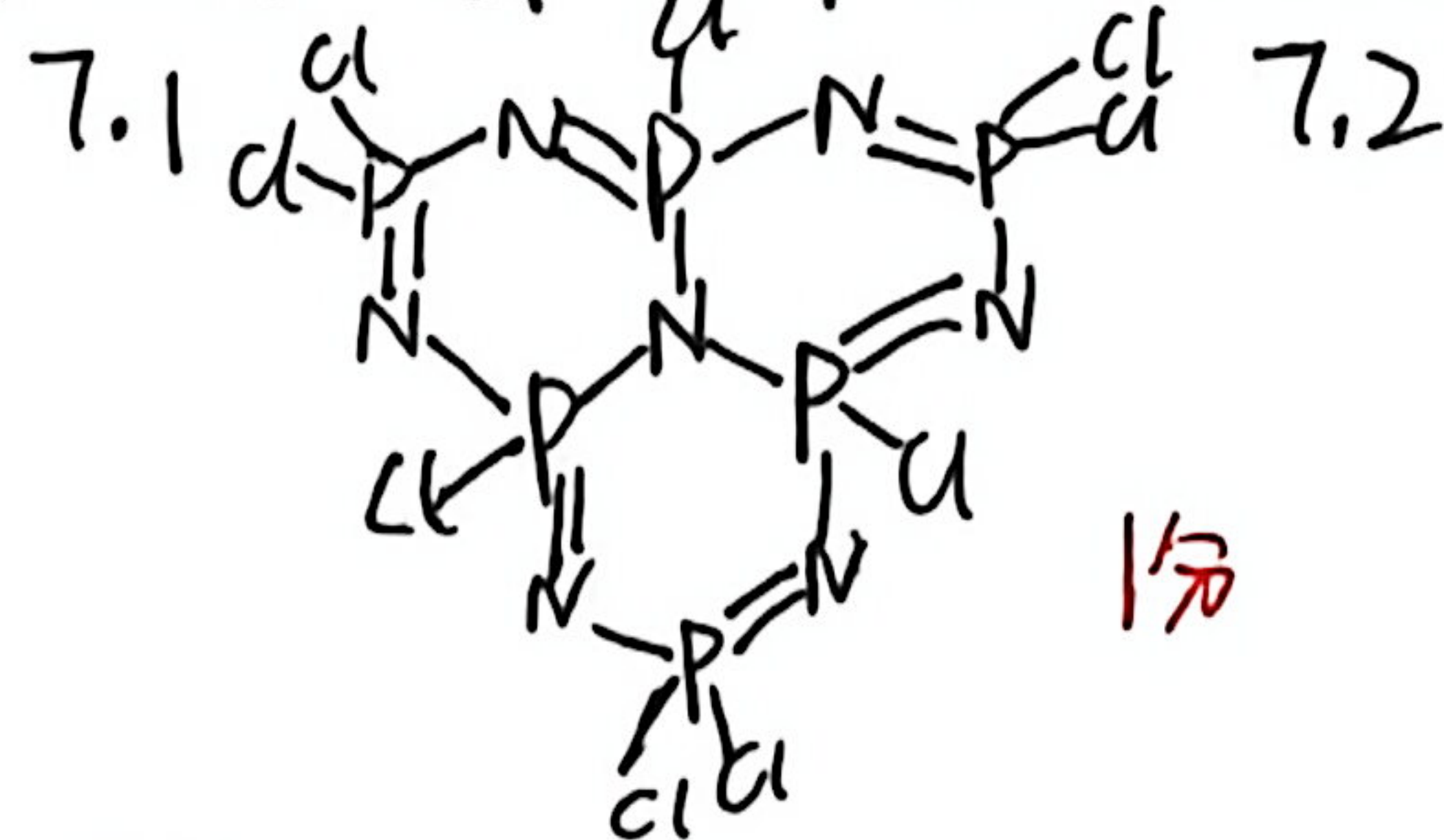
6.1 KNO_3 石硝 各1分, 共2分

6.2 NO_2 1分

6.3 A: 氧化连二次硝酸钠(低亚硝酸) B: 连二次硝酸根钠 各1分, 共2分



第7题 磷的日常 (17分, 折合加8分)



第8题 思维, 在这里成长 (6分, 折合加7分)

8.1 氧化态为+1 4 4 eg 2分 (分裂图1分, 填入1分)

1分

4 4 4 4

8.2 氧化态为+3 1分

8.3 矛盾, 实际组成为 NO^- 和 Fe^{3+} 。磁矩信息是因为 Fe^{3+} 高自旋 ($S=5/2$) 与 NO^- ($S=1$) 的反铁磁性耦合的结果 (产生 $S=3/2$)

宋天佑教材代表了经典简化的教学模型, 而 Housecroft 则为现代无机化学对于谱学证据的精确认识 2分

第9题 (1分, 折合加2分)

